2.3 Theorems about Perpendicular Lines

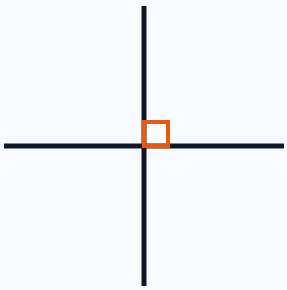
Perpendicularity is a stronger statement than just "one right angle" — it forces all four angles at the intersection to be $90°$, which is what lets a single right-angle mark unlock an entire figure. This section links perpendicularity to parallelism in two directions: two lines each perpendicular to a third must be parallel to each other, and a line perpendicular to one of two parallel lines is automatically perpendicular to the other (the Perpendicular Transversal Theorem). Expect a short proof, not just a fact to quote, along with algebra where the right-angle condition itself becomes an equation.

Examples

Example 1: Definition of perpendicular

If $\ell_1 \perp \ell_2$, what angle measure is formed, and how many angles have that measure?

Solution



Perpendicular lines mark a right angle with a small square, not an arc.

By definition, perpendicular lines intersect to form right angles — each exactly $90°$.

Saying $\ell_1 \perp \ell_2$ isn't just about one angle: it guarantees *all four* angles at the intersection are $90°$, not only the one drawn.

Answer: $90°$, and all four angles at the intersection equal it.

Example 2: Why all four angles must be right

Two lines are perpendicular. Explain, starting from just one $90°$ angle, why the other three angles at the intersection must also be $90°$.

Solution

Start from one right angle ($90°$) at the intersection.

Its linear-pair partner is supplementary: $180° - 90° = 90°$ — also right.

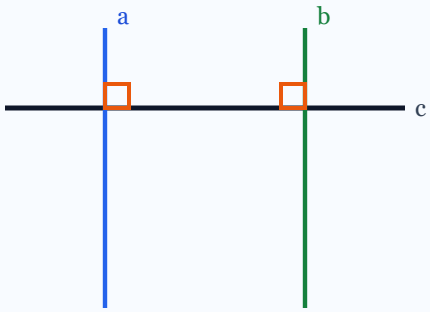
Its vertical angle is congruent to it (Vertical Angles Theorem), so it is also $90°$; the fourth angle, vertical to the second, is $90°$ as well.

Answer: All four angles measure $90°$ — it only takes knowing *one* of them is right to force perpendicularity.

Example 3: Two perpendiculars to the same line

If $\ell_1 \perp t$ and $\ell_2 \perp t$, what can you conclude about ℓ_1 and ℓ_2?

Solution



Lines a and b are both perpendicular to c — they never meet.

Both intersections with t create $90°$ angles in corresponding position (same side of t, same corner).

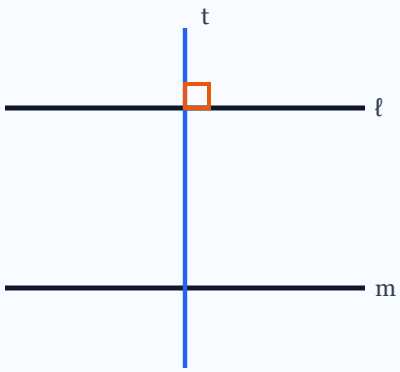
Since those corresponding angles are congruent ($90° = 90°$), the Converse of the Corresponding Angles Postulate applies.

Answer: $\ell_1 \parallel \ell_2$.

Example 4: Perpendicular Transversal Theorem

$\ell \parallel m$ and $t \perp \ell$. What can you conclude about t and m?

Solution



t is perpendicular to ℓ; since ℓ ∥ m, it must be perpendicular to m too.

Perpendicular Transversal Theorem: if a line is perpendicular to one of two parallel lines, it is automatically perpendicular to the other.

Here $t \perp \ell$ and $\ell \parallel m$ satisfy exactly the theorem's hypothesis.

Answer: $t \perp m$.

Example 5: Any labeled angle is 90°

Two lines are perpendicular. One of the four angles at the intersection is labeled x. Find x.

Solution

It doesn't matter which of the four angles is labeled — perpendicularity forces every one of them to be a right angle.

Answer: $x = 90$.

Example 6: Linear pair with a right angle

One angle at an intersection is $90°$. Find the measure of the angle adjacent to it (its linear-pair partner).

Solution

Linear Pair Postulate: adjacent angles whose outer sides form a line are supplementary.

$180° - 90° = 90°$.

Answer: $90°$ — confirming that a linear pair including one right angle always produces another right angle.

Example 7: Horizontal and vertical lines

On a coordinate plane, why are a horizontal line and a vertical line always perpendicular, even though the usual "slopes multiply to -1" rule doesn't directly apply?

Solution

The product rule ($m_1 m_2 = -1$) fails here because a vertical line's slope is undefined — you can't multiply by "undefined."

Instead, use the definition directly: a horizontal line makes a $0°$ angle with the x-axis, and a vertical line makes a $90°$ angle with it, so the angle between the two lines themselves is $90°$.

Answer: Horizontal and vertical lines are always perpendicular — this is the one case handled by definition rather than the slope-product rule (see Section 2.4).

Example 8: Reading the right-angle mark

A diagram shows a small square drawn at an intersection instead of a curved arc. What does that convention mean, and why is it different from a general angle mark?

Solution

A curved arc marks a generic angle of unspecified measure; a small square specifically marks a $90°$ angle — it is a promise, not just a label.

Seeing the square lets you use $90°$ immediately in any equation, without being told the measure in words.

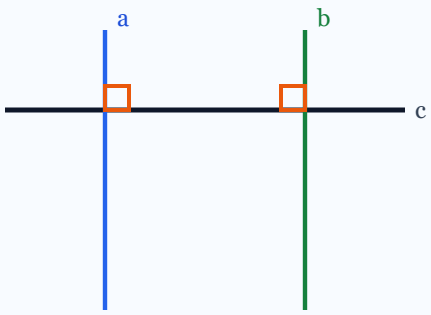
Answer: The square symbol means the angle is exactly $90°$ (the lines are perpendicular there).

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Example 9: A short proof that two perpendiculars are parallel

Given: $a \perp c$ and $b \perp c$. Prove: $a \parallel b$.

Solution



Both a and b meet c at 90°, in matching corresponding positions.

1. $a \perp c$ and $b \perp c$ — *Given*
2. The angle formed by a, c is $90°$; the corresponding angle formed by b, c is $90°$ — *Definition of perpendicular*
3. $90° = 90°$, so the two corresponding angles are congruent — *Substitution*
4. $a \parallel b$ — *Converse of the Corresponding Angles Postulate*

■ **Proved:** $a \parallel b$.

Example 10: Algebra inside a right-angle condition

Transversal t meets line ℓ, forming an angle of $(2x + 10)°$ that is marked as a right angle. Given $\ell \parallel m$, find x, and explain why t must also be perpendicular to m.

Solution

The right-angle mark tells us the expression itself equals $90°$: $2x + 10 = 90$.

$2x = 80 \Rightarrow x = 40$.

Since $t \perp \ell$ and $\ell \parallel m$, the Perpendicular Transversal Theorem guarantees $t \perp m$ as well.

Answer: $x = 40$; $t \perp m$.

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Definition

$\ell \perp m$ means they form what measure?

Answer: **Verification:**

Problem 2: Count right angles

Perpendicular lines form how many right angles?

Answer: **Verification:**

Problem 3: Two perps

$a \perp c$ and $b \perp c$. Then a and b are?

Answer: **Verification:**

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Problem 4: Perp transversal

$\ell_1 \parallel \ell_2$ and $t \perp \ell_1$. Then t and ℓ_2?

Answer: **Verification:**

Problem 5: Measure

Perpendicular intersection; one angle is?

Answer: **Verification:**

Problem 6: Adjacent

Right angle's linear-pair partner?

Answer: **Verification:**

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Problem 7: Horiz/vert

Horizontal line and vertical line are always?

Answer: **Verification:**

Problem 8: Symbol

Symbol for perpendicular?

Answer: **Verification:**

Problem 9: Parallel symbol

Symbol for parallel?

Answer: **Verification:**

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Problem 10: Conclusion

Two lines each ⊥ to same line. They are?

Answer: **Verification:**

Problem 11: Find x

Perpendicular lines; one angle measures x. Find x.

Answer: **Verification:**

Problem 12: Transversal

Parallel lines; transversal ⊥ one. It is ⊥ the other?

Answer: **Verification:**

Problem 13: Not parallel?

Can two distinct lines both $\perp$ the same line and also intersect each other?

Answer: **Verification:**

Problem 14: Challenge — explain the logic

Lines a and b are both perpendicular to line c. Explain in one sentence, using corresponding angles, why a must be parallel to b.

Answer: **Verification:**

Problem 15: Challenge — algebra with a right angle

Transversal t meets ℓ at an angle marked $(4x - 6)^\circ$ as a right angle, and $\ell \parallel m$. Find x, and state the measure of the angle t forms with m.

Answer: **Verification:**

✓ Answer Key

1. 90°
2. 4
3. Parallel
4. Perpendicular
5. 90°
6. 90°
7. Perpendicular
8. $\perp$
9. $\parallel$
10. Parallel
11. $x = 90$
12. Yes
13. No
14. Both a and b meet c at 90° in matching corresponding positions, so those corresponding angles are congruent, and by the Converse of the Corresponding Angles Postulate $a \parallel b$
15. $x = 24$; t meets m at 90°

Great work — keep practising!